

Notes: Chen, Gentry, Li, and Lu (RES, 2026)

Identification and Inference in First-Price Auctions with Risk-Averse Bidders and Selective Entry

Contents

1	Big picture	3
2	Data and measurement (observable and latent key objects)	4
2.1	What the econometrician observes	4
2.2	Timing and primitives of the AS-RA model	4
2.3	Entry thresholds and post-entry value distributions	4
2.4	Bid distributions and quantile objects	5
2.5	The utility transformation	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Equilibrium bidding with selective entry	5
3.2	Boundary conditions: the novel twist	6
3.3	Inverse bidding in quantile form	6
3.4	Candidate latent quantiles and the sharp identified set	6
3.5	Theorem 1: restrictions M, O, I, D, and S	6
3.6	Why the theorem is powerful	7
3.7	Risk neutrality is nonparametrically testable	7
3.8	Continuous cost-shifter variation and local curvature	8
3.9	Identification with variation in both N and z	8
3.10	Why there is only identification up to a constant	8
3.11	Point identification under parametric utility	8
3.12	Conditional identification with a parametric copula	9
3.13	Flexible parametric likelihood framework	9
3.14	Likelihood and bid density	10
3.15	Inference with likelihood-ratio confidence sets	10

3.16	MPEC implementation	10
3.17	Simulation design	11
4	Identification and instruments (what makes the estimates credible)	11
4.1	Why standard risk-aversion identification fails with selective entry	11
4.2	What identifies risk aversion here	12
4.3	Role of the exclusion restrictions	12
4.4	Why top bids matter so much	12
4.5	Why reserve-price boundary conditions matter for credibility	12
4.6	Why inference is difficult	13
4.7	Relation to the literature	13
5	Tables and figures	13
5.1	Figure 1: recursive identification with three competition structures	13
5.2	Figure 2: pointwise KL contour sets	13
5.3	Figure 3: Frank copula versus flexible copula	14
5.4	Figure 4: posterior and frequentist LR distributions	14
5.5	Figure 5 and Table 1: point-identified inference under the Frank copula model . . .	14
5.6	Table 2: flexible copula point-identified inference	16
5.7	Figures 6 and 7: set-identified coverage for $Q = 4$ and $Q = 6$	16
6	Short summary	17
7	Conclusion	18
7.1	Core conclusion (what the paper establishes)	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** How can we identify bidder risk aversion in first-price auctions when bidders decide *whether to enter* before bidding, and how can we do valid inference when the model may be only set identified and the support of observed bids depends on the parameters?
- **Setting.** The paper studies a first-price auction with risk-averse bidders and selective entry. Potential bidders first observe a private signal about their value, then decide whether to pay an entry cost, and only entrants bid. The paper calls this flexible environment the **Affiliated Signal with Risk Aversion (AS-RA)** model.
- **Core challenge.** Standard identification results for bidder risk aversion typically rely on the distribution of entrants' values being invariant across auction environments. With *selective entry*, that invariance fails: changing the number of potential bidders or the entry-cost shifter changes who enters and therefore changes the distribution of values among entrants.
- **Additional challenge.** The paper allows for *weakly non-binding reserve prices*. This matters because the lower boundary condition of the equilibrium bidding differential equation depends on whether entry is partial or complete. With full entry, the lowest type bids her value; with partial entry, the lowest entrant bids the reserve price.
- **Main theoretical result.** The paper gives a *sharp nonparametric characterization* of the bid-stage identified set for utility, the ex ante value distribution, and the value-signal copula. A candidate inverse utility function is rationalizable if and only if the implied latent value quantiles satisfy five restrictions, labelled **M, O, I, D, and S**.
- **Main identification result.** Risk neutrality is *nonparametrically testable*. With sufficient variation in the number of potential bidders N and an entry-cost shifter z , the AS-RA primitives are identified up to a bounded constant on their equilibrium domain. Under CRRA or CARA utility, point identification is possible.
- **Main inference result.** The paper adapts and extends the likelihood-ratio confidence-set methods of Chen, Christensen, and Tamer (2018) to this auction environment. The key extension is to make subvector inference valid in set-identified models with *parameter-dependent support*, which is exactly what happens in first-price auctions because structural parameters move the maximum bid support.
- **Computational contribution.** The paper develops fast MPEC-based methods, based on Chebyshev collocation and a stable evaluation of bid densities, to estimate and profile flexible auction models without closed-form bidding functions.
- **Why this matters.** The paper shows that once entry is endogenous, risk-aversion identification cannot just import the GPV or CGPV logic. Entry selection and boundary conditions are first-order econometric objects, not technical details.
- **Economic takeaway.** In auctions with selective entry, what the econometrician can recover from bids is not the primitive value distribution directly but a collection of post-entry latent quantile functions tied together across auction environments by equilibrium restrictions. Those cross-environment restrictions are what identify risk aversion.

2 Data and measurement (observable and latent key objects)

2.1 What the econometrician observes

- For each auction, the econometrician observes:
 - the number of potential bidders N ,
 - the number of entrants and their bids,
 - the reserve price p_0 ,
 - and, in some settings, a cost shifter z that affects the opportunity cost of entry.
- The number of potential bidders varies over a finite set

$$\mathcal{N} = \{N_1, \dots, N_K\}, \quad N_1 < N_2 < \dots < N_K.$$

- If an instrument exists, it can be either discrete or continuous. The especially powerful case for identification is a continuous z that shifts entry costs but leaves bid-stage primitives unchanged.
- The econometrician does *not* observe the private signals, the private values, the utility function, or the value-signal copula.

2.2 Timing and primitives of the AS-RA model

- There are N symmetric potential bidders.
- Each bidder has a value-signal pair (V_i, S_i) , with signals normalized so that $S_i \sim U[0, 1]$.
- The joint distribution of values and signals is represented by

$$F_{VS}(v, s) = C(F(v), s),$$

where:

- F is the ex ante value distribution,
- C is a copula linking values and signals,
- the signal is affiliated with the value in the sense encoded by the copula restrictions.
- Bidders have smooth concave Bernoulli utility $U(\cdot)$, so the paper allows for risk aversion.
- The reserve price is only assumed to be *weakly non-binding*: it may satisfy $p_0 \leq v$, where v is the lower support of values.

2.3 Entry thresholds and post-entry value distributions

- At competition level k and cost shifter z , entry is characterized by a threshold rule:

$$S_i \geq s_k(z) \iff \text{bidder } i \text{ enters.}$$

- Entry thresholds are identified from observed entry behavior because signals are uniform on $[0, 1]$.

- Given an entry threshold $s_k(z)$, the distribution of values among entrants is

$$F_k^0(v | z) = F_0(v | S_i \geq s_k(z)) = \frac{F_0(v) - C_0(F_0(v), s_k(z))}{1 - s_k(z)}.$$

- This is the main source of difficulty relative to exogenous-entry models: the entrant-value distribution changes with N and z through the endogenous entry threshold $s_k(z)$.

2.4 Bid distributions and quantile objects

- Let $G_k(\cdot | z)$ denote the equilibrium bid distribution, with density $g_k(\cdot | z)$.
- Let $b_k(\alpha | z)$ denote the bid quantile function and $v_k(\alpha | z)$ the latent value quantile function among entrants.
- Much of the paper works in quantile space because quantile inverse bidding makes the relationship between observables and risk aversion especially transparent.

2.5 The utility transformation

- Define

$$\lambda_0(x) \equiv \frac{U_0(x)}{U'_0(x)}.$$

- This is the natural object entering the bidding first-order condition.
- Under weak risk aversion, $U''_0 \leq 0$, one has

$$\lambda'_0(x) \geq 1 \quad \implies \quad 0 \leq (\lambda_0^{-1})'(r) \leq 1.$$

- Under global risk neutrality, $\lambda_0(x) = x$ and therefore

$$\lambda_0^{-1}(r) = r, \quad (\lambda_0^{-1})'(r) = 1.$$

3 Models and empirical specifications (all equations + interpretation)

3.1 Equilibrium bidding with selective entry

Conditional on entry threshold $s_k(z)$ and post-entry value distribution $F_k^0(\cdot | z)$, a bidder with value v_i solves

$$\max_y U_0(v_i - \beta_k(y | z)) \left[s_k(z) + (1 - s_k(z)) F_k^0(y | z) \right]^{N_k - 1}.$$

The equilibrium first-order condition implies the bidding differential equation

$$\beta'_k(v | z) = \lambda_0(v - \beta_k(v | z)) \frac{(N_k - 1)(1 - s_k(z)) f_k^0(v | z)}{s_k(z) + (1 - s_k(z)) F_k^0(v | z)}.$$

Interpretation. This is the GPV/CGPV style bidding equation, but now the probability of beating a rival depends both on the entrant-value distribution and on the possibility that rivals do not enter. Selective entry therefore enters the bidding equation multiplicatively and changes identification.

3.2 Boundary conditions: the novel twist

The bidding differential equation must be paired with a lower boundary condition, and this boundary depends on whether entry is full or partial.

- If $s_k(z) = 0$ so that all bidders enter, the lowest type bids her value:

$$\beta_k(v \mid z) = v.$$

- If instead $s_k(z) \in (0, 1)$ so that entry is partial, the lowest *entrant* wins with positive probability even by bidding the reserve price, so the correct boundary is

$$\beta_k(v_k(0 \mid z) \mid z) = p_0.$$

Interpretation. This difference is a central contribution of the paper. The econometric boundary condition changes discontinuously when moving from complete to incomplete entry. Ignoring this would contaminate inverse bidding and any identification argument built on it.

3.3 Inverse bidding in quantile form

The paper rewrites the bidding equation in quantile space. Define

$$R_k(\alpha \mid z) \equiv \frac{s_k(z) + (1 - s_k(z))\alpha}{(N_k - 1)(1 - s_k(z))} b'_k(\alpha \mid z).$$

Then the latent entrant-value quantile function satisfies

$$v_k(\alpha \mid z) = b_k(\alpha \mid z) + \lambda_0^{-1}(R_k(\alpha \mid z)), \quad \alpha \in [0, 1].$$

Interpretation. Bids and entry data identify $b_k(\alpha \mid z)$ and $R_k(\alpha \mid z)$. The only unknown part is the inverse utility object λ_0^{-1} . Thus, for any candidate λ^{-1} , the econometrician can construct a candidate latent value quantile function.

3.4 Candidate latent quantiles and the sharp identified set

For any candidate inverse utility function λ^{-1} , define the candidate latent quantile function

$$\tilde{v}_k(\alpha \mid z; \lambda^{-1}) \equiv b_k(\alpha \mid z) + \lambda^{-1}(R_k(\alpha \mid z)).$$

The main theorem says that a candidate λ^{-1} is observationally consistent with the bid data if and only if the implied candidate quantiles satisfy five restrictions.

3.5 Theorem 1: restrictions M, O, I, D, and S

- **M (Monotonicity).** For every competition level and shifter, the candidate quantile function must be strictly increasing. Equivalently, the implied entrant-value density must stay positive.
- **O (Ordering).** If one auction environment has a lower entry threshold than another, then its entrant-value quantiles must be weakly lower. When thresholds are equal, the candidate quantiles must coincide.

- **I (Invariant top quantile).** The highest possible entrant value must be the same across all competition levels and instruments:

$$\tilde{v}_k(1 \mid z; \lambda^{-1}) = \tilde{v}_l(1 \mid z'; \lambda^{-1}) \quad \forall k, l, z, z'.$$

- **D (Density ordering).** Conditional densities induced by different thresholds must be compatible with a common underlying ex ante value distribution and copula.
- **S (Shape / slope consistency).** The family of conditional entrant-value distributions across thresholds must satisfy a cross-threshold convexity-type restriction implied by a single common copula.

Interpretation. The first three restrictions are the easiest to understand economically: monotonicity, stochastic ordering with tougher entry, and a common upper support. The last two are the harder cross-environment consistency conditions that make the theorem *sharp*. Together they are necessary and sufficient for a common (F, C) to exist behind the observed bids.

3.6 Why the theorem is powerful

Theorem 1 shows that the bid-stage identified set can be written as

$$\mathcal{I} = \{(\lambda^{-1}, F, C) : \text{the candidate quantiles } \tilde{v}_k \text{ satisfy M,O,I,D,S}\}.$$

In other words, once the paper transforms bids into candidate latent quantiles, identification becomes a problem of checking whether these quantiles can be generated by one common ex ante environment.

3.7 Risk neutrality is nonparametrically testable

Restriction I implies that for any two auction environments,

$$b_l(1 \mid z') - b_k(1 \mid z) = \lambda^{-1}(R_k(1 \mid z)) - \lambda^{-1}(R_l(1 \mid z')).$$

If bidders are risk neutral, then $\lambda^{-1}(r) = r$ over the relevant equilibrium domain. Since weak risk aversion implies $(\lambda^{-1})' \leq 1$, any strict concavity of utility compresses differences in top bids relative to the risk-neutral benchmark.

The paper's corollary says:

- if bidders are risk neutral, any rationalizing λ^{-1} must be linear with slope 1 over the observed equilibrium domain;
- if bidders are strictly risk averse anywhere on that domain, no risk-neutral model can rationalize the observed bidding behavior.

Interpretation. This is stronger than saying risk aversion is point identified under special assumptions. The paper shows that risk neutrality itself can be rejected nonparametrically.

3.8 Continuous cost-shifter variation and local curvature

When z varies continuously and shifts entry costs while leaving bid-stage primitives fixed, the paper shows that one can identify derivatives of the inverse utility object over an interval:

$$(\lambda_0^{-1})'(r) \text{ is identified on an interval generated by } R_k(1 | z).$$

This in turn identifies Arrow-Pratt absolute risk aversion on that interval.

Interpretation. Continuous variation in entry costs effectively traces out how the same upper-support restriction behaves at different arguments of λ^{-1} , which turns a level restriction into a slope restriction.

3.9 Identification with variation in both N and z

Suppose there are three competition structures (N_k, z_k) , (N_l, z_l) , and (N_m, z_m) such that they induce the same entry threshold:

$$s_k(z_k) = s_l(z_l) = s_m(z_m).$$

Because the entry threshold is the same, Restriction O implies the same latent entrant-value distribution across those environments. The paper then recursively matches values of the observable objects R_k , R_l , and R_m across quantiles. This recursion is illustrated in Figure 1.

Theorem 2 shows that on the equilibrium domain,

$$\lambda_0^{-1}(r) = \phi_1(r) + \phi_2,$$

where $\phi_1(r)$ is identified and ϕ_2 is an additive constant belonging to a bounded interval.

Interpretation. Variation in both the number of potential bidders and the entry-cost shifter lets the econometrician compare different auctions that have the same selection cutoff but different competitive pressure. That is what identifies the shape of λ^{-1} .

3.10 Why there is only identification up to a constant

The additive constant appears because, with a weakly non-binding reserve, the lower boundary of the latent value support need not be directly pinned down by bids. If the reserve is *just-binding*, then the constant vanishes. This is why the paper emphasizes boundary conditions so strongly.

Interpretation. The unobserved lower boundary is the only remaining degree of freedom once enough cross-environment variation exists. If the reserve just binds, that degree of freedom disappears.

3.11 Point identification under parametric utility

Suppose utility belongs to a parametric family, with parameter γ . Restriction I implies

$$\bar{v} = b_k(1 | z) + \lambda^{-1}(R_k(1 | z); \gamma) \quad \forall k, z,$$

where $\bar{v}$ is the common upper support of entrant values.

The paper proves that if utility is either:

- **CRRA:** $U(x) = x^{1-\rho}$,
- or **CARA:** $U(x) = x$ when $\gamma = 0$ and $U(x) \propto 1 - e^{-\gamma x}$ otherwise,

then enough variation in $R_k(1 \mid z)$ yields point identification of λ_0^{-1} and of the post-entry distributions $F_1^0, \dots, F_K^0$.

Interpretation. Once the shape of λ^{-1} is pinned down by a low-dimensional parameter, the invariant-top-quantile equations are enough to solve for the parameter itself.

3.12 Conditional identification with a parametric copula

If instead the copula is parameterized as $C(a, s; \theta)$, define

$$h_k(a; \theta) = \frac{a - C(a, s_k; \theta)}{1 - s_k}.$$

For the correct copula parameter θ_0 , one has

$$F_k^0(y) = h_k(F_0(y); \theta_0).$$

This yields compatibility equations across competition levels,

$$b_k(h_k(a; \theta)) + \lambda_0^{-1}(R_k(h_k(a; \theta))) = b_l(h_l(a; \theta)) + \lambda_0^{-1}(R_l(h_l(a; \theta))),$$

which the paper uses to identify λ^{-1} up to an additive constant conditional on θ .

Interpretation. A parametric copula restricts how selection changes the entrant-value distribution across competition levels. Those restrictions replace some of the work otherwise done by observing a continuous instrument.

3.13 Flexible parametric likelihood framework

For inference, the paper focuses on the no- z case and flexibly parameterizes the bid-stage primitives.

First, utility is parameterized through λ as

$$\lambda(x) = \bar{v} \tilde{\lambda}(\bar{v}^{-1}x \mid \gamma), \quad \tilde{\lambda}(u \mid \gamma) = u + \sum_{j=1}^Q \gamma_j B_{j,Q}(u),$$

where $B_{j,Q}(u)$ are Bernstein basis polynomials. Monotonicity and risk-aversion-consistency are enforced by constraints on the coefficients γ .

Second, each post-entry value distribution is parameterized as a Bernstein polynomial on $[v, \bar{v}]$:

$$F_k(y) = \tilde{F}_k\left(\frac{y - v}{\bar{v} - v} \mid \phi_k\right), \quad \tilde{F}_k(u \mid \phi_k) = \sum_{j=1}^{P-1} \phi_{k,j} B_{j,P}(u) + B_{P,P}(u).$$

The coefficient vectors are constrained so that the implied distributions satisfy the translated versions of the theorem's restrictions: M', O', D', and S'.

The full parameter vector is

$$\psi = (\bar{v}, v, s, \gamma, \phi), \quad \dim(\psi) = 2 + K + Q + K(P - 1).$$

Interpretation. The paper turns a nonparametric identified-set problem into a flexible finite-dimensional estimation problem, while preserving the economic restrictions implied by theory.

3.14 Likelihood and bid density

For any candidate parameter vector, the model-implied bid density at competition level k is

$$g_k(b \mid \psi) = f_k(\beta_k^{-1}(b \mid \psi) \mid \bar{v}, v, \phi_k) \cdot \beta_k^{-1'}(b \mid \psi).$$

Using the bidding differential equation, the paper obtains a more stable computational expression:

$$g_k(b \mid \psi) = \frac{s_k + (1 - s_k)F_k(v \mid \bar{v}, v, \phi_k)}{\lambda(v - b \mid \bar{v}, \gamma)(N_k - 1)(1 - s_k)},$$

where v is the latent value satisfying $b = \beta_k(v \mid \psi)$.

If $e_{il} \in \{0, 1\}$ indicates whether potential bidder i enters auction l , the log-likelihood contribution is

$$\ell_{il}(\psi) = \log \left[s_{k_l}^{1-e_{il}} \left((1 - s_{k_l}) g_{k_l}(b_{il} \mid \psi) \right)^{e_{il}} \right].$$

Interpretation. The same parameter vector must explain both entry rates and the entire bid distribution. That joint fit is what makes the structural model empirically meaningful.

3.15 Inference with likelihood-ratio confidence sets

Let

$$Q(\psi) = 2[L(\hat{\psi}) - L(\psi)]$$

be the likelihood-ratio statistic.

- **Procedure 1** uses posterior draws of $Q(\psi)$ to build a confidence set for the full identified set of ψ .
- **Procedure 2** profiles the likelihood over all parameter vectors observationally equivalent for a lower-dimensional object $\eta(\psi)$ and builds a confidence set for that subvector or functional.

A key complication is that auction models have *parameter-dependent support*: changing structural parameters changes the support of bids, especially the maximum bid. The appendix extends the Chen-Christensen-Tamer theory so that Procedure 2 remains valid in this setting.

Interpretation. This is one of the paper's biggest econometric contributions. Standard likelihood asymptotics are fragile when support depends on the parameter, but that is exactly the first-price-auction case.

3.16 MPEC implementation

The paper does not solve equilibrium bid functions by directly integrating the ODE each time inside optimization. Instead, it:

- represents equilibrium bidding functions with Chebyshev collocation,
- treats the collocation conditions as constraints in an MPEC problem,
- partially discretizes low bids while evaluating high bids exactly,

- and uses this to solve both MLE and profile-likelihood problems quickly and stably.

Interpretation. Without these numerical tricks, the likelihood-based procedures would be too slow to be practical in flexible auction models.

3.17 Simulation design

The simulation section uses a transparent benchmark DGP:

- **Utility:** CRRA, so $U_0(x) = x^{1-\rho_0}$ and $\lambda_0(x) = x/(1 - \rho_0)$.
- **Value distribution:** truncated logistic with mean 0.5 and scale 0.3 on $[0, 1]$.
- **Reserve:** $p_0 = 0$, and for the simulation exercise the econometrician knows the reserve is just-binding, so the lower support equals the reserve.
- **Copula:** Frank copula with parameter $\theta_0 = 2.0$, implying Spearman rank correlation of about 0.32.
- **Entry cost:** $c_0 = 0.1$.
- **Competition levels:** $N \in \{2, 4, 6, 8\}$.
- **DGPs:** one risk-averse DGP with $\rho_0 = 0.5$ and one risk-neutral DGP with $\rho_0 = 0$.
- **Sample sizes:** average bids per competition level $M \in \{1000, 2000\}$.

The paper compares two econometric specifications:

1. a **Flexible copula** model, where the post-entry value distributions are flexibly parameterized;
2. a **Frank copula** model, where the copula is correctly parameterized and only the value distribution is approximated flexibly.

The main object for inference is the average slope

$$\bar{\lambda}_0 = \frac{\tilde{\lambda}(x \mid \gamma_0)}{x},$$

which under the CRRA DGP equals $1/(1 - \rho_0)$.

4 Identification and instruments (what makes the estimates credible)

4.1 Why standard risk-aversion identification fails with selective entry

- In GPV-style models with exogenous entry, changing the number of competitors changes the bidding environment but not the latent value distribution of bidders.
- With selective entry, a different number of potential bidders or a different entry-cost shifter changes who enters.
- Therefore, the distribution of values among entrants is itself endogenous to the auction environment. This breaks the key invariance used by standard nonparametric identification arguments.

4.2 What identifies risk aversion here

- Observed bids and entry rates identify bid quantiles $b_k(\alpha | z)$ and the objects $R_k(\alpha | z)$.
- A candidate inverse utility function then implies candidate latent value quantiles.
- Cross-auction restrictions M, O, I, D, and S determine whether those candidate quantiles can all come from one common ex ante environment.

Key point. Identification comes from the requirement that all observed auction environments be simultaneously rationalizable by one utility function and one ex ante value-signal structure.

4.3 Role of the exclusion restrictions

The identifying exclusion restriction is that N and the cost shifter z affect entry conditions but do *not* directly change bid-stage primitives (U_0, F_0, C_0) . This is what makes cross-environment comparisons informative.

- Changing z shifts the opportunity cost of entry, which moves the threshold $s_k(z)$.
- Changing N changes the intensity of competition and, through equilibrium, the observed bid distribution.
- Because the primitive bid-stage environment is held fixed, those observed changes must be absorbed through equilibrium restrictions rather than through arbitrary shifts in primitives.

4.4 Why top bids matter so much

Restriction I uses the invariant top quantile across auction environments. That makes the upper support of bids especially informative.

- In the continuous- z case, moving z traces out the derivative of λ^{-1} through changes in top bids.
- In the flexible-copula likelihood, identification is consequently strongest near the upper tail of the bid distribution.

Interpretation. The paper's inference results later confirm this: in weakly identified flexible models, precision comes largely from bids near the maximum.

4.5 Why reserve-price boundary conditions matter for credibility

A major contribution of the paper is showing that incomplete entry changes the relevant lower boundary.

- If one incorrectly imposed the full-entry boundary in a partial-entry environment, the inverse bidding map would be misspecified.
- That misspecification would show up exactly where identification is already delicate: at the bottom of the entrant-value support and in the additive-constant issue of Theorem 2.

Interpretation. The boundary condition is not a cosmetic modeling choice. It is structurally tied to whether the model is identified only up to a constant or fully point identified.

4.6 Why inference is difficult

- The model may be set identified rather than point identified.
- The support of bids depends on the parameters.
- Equilibrium bidding functions generally have no closed form.
- Profile likelihood requires repeatedly solving equilibrium and checking observational equivalence.

Interpretation. This combination explains why the paper devotes so much effort to MPEC computation and to extending the CCT theory.

4.7 Relation to the literature

- Relative to **GPV** and **CGPV**, the paper allows endogenous entry and shows how this changes risk-aversion identification.
- Relative to **LLZ** and **GL**, it adds bidder risk aversion and non-binding reserve prices.
- Relative to the broader partial-identification literature, it provides a sharp characterization of an equilibrium identified set in a flexible auction environment.
- Relative to **Chen, Christensen, and Tamer (2018)**, it extends subvector confidence-set methods to models with parameter-dependent support.

5 Tables and figures

5.1 Figure 1: recursive identification with three competition structures

Read:

- Figure 1 visualizes the recursion behind Theorem 2. The idea is to take three auction environments with the same entry threshold but different numbers of potential bidders.
- Matching the observable R_k , R_l , and R_m across these environments lets the econometrician recursively pin down λ^{-1} over its equilibrium domain, up to a possible additive constant.
- The figure makes the additive-constant issue intuitive: if the lower boundary is not pinned down by a just-binding reserve, the recursion still leaves one free constant.
- This is the cleanest visual summary of why variation in both N and z matters.

5.2 Figure 2: pointwise KL contour sets

Read:

- Figure 2 plots profiled KL contours for the average slope object $\tilde{\lambda}(x \mid \gamma_0)/x$.
- Under $Q = 1$ (the CRRA case), contours are tight and essentially point identifying.
- Under $Q = 4$ and $Q = 6$, contours flatten substantially, especially in the fully flexible copula model. This is direct evidence of set identification or weak point identification.

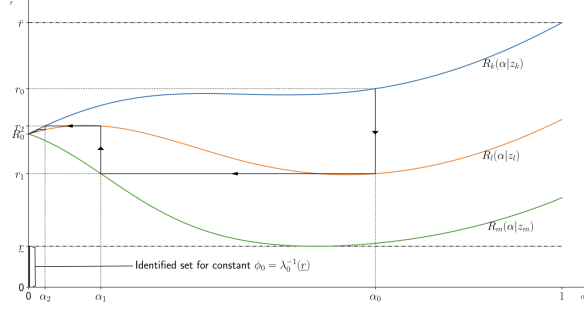


FIGURE 1
Constructing a recursive sequence to identify $\lambda_0^{-1}(r)$ based on three competition structures (N_k, z_k) , (N_l, z_l) , (N_m, z_m) such that $s_k(z_k) = s_l(z_l) = s_m(z_m)$

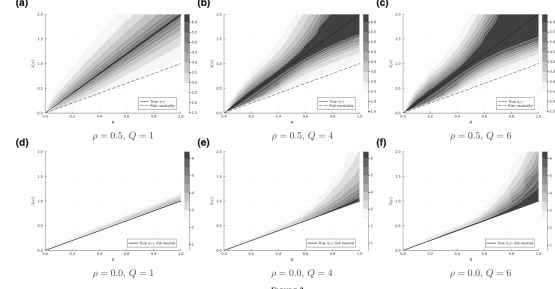


FIGURE 2
Pointwise KL contour sets for $\lambda(x|y_0) = x/(1-p_0)$, parameterizing $\lambda_0(x)$ as Bernstein of order $Q \in \{1, 4, 6\}$ and $F_1^Q, \dots, F_6^Q$ as Bernstein of order $P = 6$. (a) $\rho = 0.5, Q = 1$. (b) $\rho = 0.5, Q = 4$. (c) $\rho = 0.5, Q = 6$. (d) $\rho = 0.0, Q = 1$. (e) $\rho = 0.0, Q = 4$. (f) $\rho = 0.0, Q = 6$.
Note: Contour values reflect the inverse order of magnitude of the pointwise profiled KL divergence $\inf_{\rho \in \Psi} \{P_{\rho}^Q(y) : \lambda(x) = y\}$. For example, a contour value above 5.0 implies that there is a parameter vector $\rho \in \Psi$ with $\lambda(x) = y$ such that $P_{\rho}^Q(y) \leq 10^{-5.0}$.

- The risk-neutral and risk-averse DGPs show the same qualitative lesson: allowing richer utility shapes can quickly make the problem much less tightly identified.

5.3 Figure 3: Frank copula versus flexible copula

Read:

- Figure 3 compares the profiled KL divergence for the same average-slope object under a correctly specified Frank copula model versus a flexible-copula model.
- The Frank copula model has much steeper and closer-to-quadratic curvature.
- The flexible-copula model is much flatter, implying weaker finite-sample identification.
- Economically, the reason is that the parametric copula uses the entire shape of the bid distribution for identification, while the fully flexible model leans much more heavily on upper-tail information.

5.4 Figure 4: posterior and frequentist LR distributions

Read:

- Figure 4 shows Q-Q plots comparing posterior LR quantiles and frequentist LR quantiles to a common Gamma distribution.
- Across both point-identified and set-identified specifications, the plotted points lie close to the diagonal.
- This is important because the paper's confidence-set procedures use posterior draws of LR statistics, but the target guarantee is frequentist coverage.
- The figure provides convincing finite-sample support for the asymptotic theory.

5.5 Figure 5 and Table 1: point-identified inference under the Frank copula model

Read:

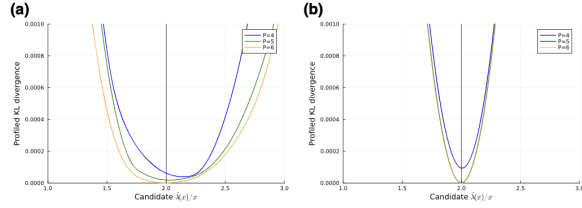


FIGURE 3
Profiled KL divergence for $\lambda_0 = 1/(1 - \rho_0)$ in risk-averse CRRA models ($\rho_0 = 0.5, Q = 1$) based on Frank copula versus Flexible copula specifications. (a) Frank: F_0 Bernstein- P , C Frank and (b) Flexible: $F_1, \dots, F_K$ Bernstein- P

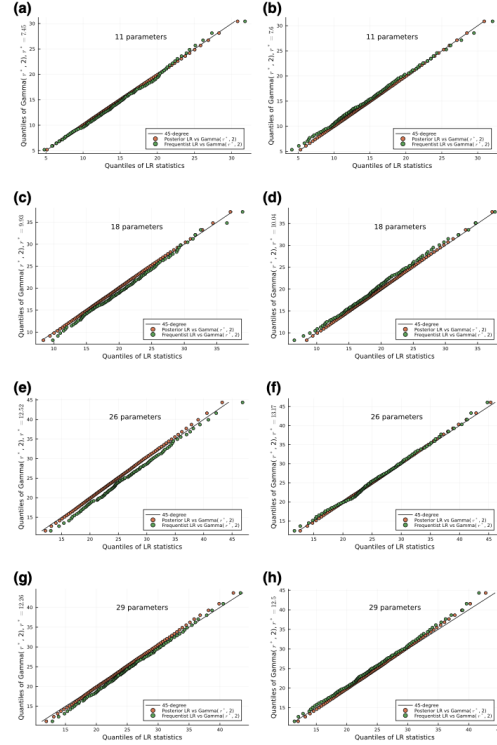


FIGURE 4
 $Q-Q$ plots of simulated posterior and frequentist LR quantiles. (a) Frank $C, \rho_0 = 0.5, P = 5, Q = 1$. (b) Frank $C, \rho_0 = 0.0, P = 5, Q = 1$. (c) Flexible $C, \rho_0 = 0.5, P = 4, Q = 1$. (d) Flexible $C, \rho_0 = 0.0, P = 4, Q = 1$. (e) Flexible $C, \rho_0 = 0.5, P = 6, Q = 1$. (f) Flexible $C, \rho_0 = 0.0, P = 6, Q = 1$. (g) Flexible $C, \rho_0 = 0.5, P = 6, Q = 4$ and (h) Flexible $C, \rho_0 = 0.0, P = 6, Q = 4$

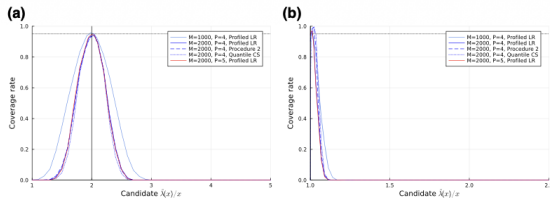


TABLE 2
Flexible copula, $Q=1$: coverage rates and median bounds based on alternative CSs for $\lambda_0 = 1/(1 - \rho_0)$,
parameterizing $\lambda_0(\cdot)$ as CRRA ($Q = 1$) and $F_1^0, \dots, F_k^0$ as Bernstein- P

Method	Prior	M	P	Risk-averse DGP $\rho_0 = 0.5, \lambda_0 = 2.0$		Risk-neutral DGP $\rho_0 = 0.0, \lambda_0 = 1.0$	
				Coverage of λ_0	Median CS bounds	Coverage (see note)	Median CS bounds
Profiled LR	—	1,000	4	0.978	[1.440, 2.795]	0.988	[1.000, 1.119]
		2,000	4	0.986	[1.569, 2.640]	0.990	[1.000, 1.098]
			5	0.990	[1.565, 2.820]	0.990	[1.000, 1.117]
			6	0.980	[1.493, 2.841]	0.990	[1.000, 1.138]
		10,000	6	0.950	[1.708, 2.503]	0.968	[1.000, 1.063]
Quantile CS	Uniform	20,000	6	0.956	[1.737, 2.355]	0.965	[1.000, 1.051]
		1,000	4	0.962	[1.659, 2.659]	0.842	[1.002, 1.127]
		2,000	4	0.954	[1.727, 2.508]	0.907	[1.001, 1.097]
			5	0.912	[1.767, 2.667]	0.854	[1.002, 1.129]
			6	0.946	[1.733, 2.658]	0.858	[1.002, 1.139]
	$\Delta_\psi = 0.01$	1,000	4	0.962	[1.527, 2.481]	0.848	[1.002, 1.125]
		2,000	4	0.966	[1.616, 2.391]	0.912	[1.001, 1.094]
			5	0.938	[1.687, 2.584]	0.827	[1.002, 1.139]
			6	0.965	[1.633, 2.530]	0.819	[1.002, 1.142]
			6	0.948	[1.583, 2.642]	0.967	[1.000, 1.110]
Procedure 2	Uniform	2,000	4	0.946	[1.689, 2.521]	0.960	[1.000, 1.085]
			5	0.946	[1.681, 2.647]	0.960	[1.000, 1.112]
			6	0.948	[1.657, 2.648]	0.957	[1.000, 1.120]
	$\Delta_\psi = 0.01$	1,000	4	0.970	[1.532, 2.668]	0.965	[1.000, 1.107]
		2,000	4	0.972	[1.629, 2.551]	0.964	[1.000, 1.083]
			5	0.954	[1.661, 2.662]	0.975	[1.000, 1.120]
			6	0.950	[1.636, 2.649]	0.961	[1.000, 1.127]

Notes: Results are based on 500 MC replications. MC CSs are based on $B^* = 400$ parameter draws $\{\mu_{b=1}^b\}_{b=1}^{400}$. For MC CSs in our risk-averse boundary DGP ($\rho_0 = 0.0$), where percentile CSs never cover λ_0 , "Coverage" reports the fraction of simulations covering $\lambda(x|\gamma_0)/x = 1.005$.

- These are the benchmark point-identified results with correctly specified copula and CRRA utility.
- In the interior risk-averse DGP, Procedure 2, profiled LR, and percentile Bayesian credible sets perform similarly.
- In the boundary risk-neutral DGP, percentile intervals perform much worse because of boundary effects, while Procedure 2 and profiled LR keep coverage near nominal levels.
- Median confidence bounds are reasonably tight around the true average slope, which shows that the paper's inference methods are practically informative in well-identified cases.

5.6 Table 2: flexible copula point-identified inference

Read:

- Table 2 repeats the exercise in the more demanding flexible-copula model with $Q = 1$.
- Profiled LR confidence sets are quite conservative at practical sample sizes, especially when compared to the nominal 95% target.
- Procedure 2 delivers coverage much closer to nominal, confirming the value of the paper's robust subvector method even when the model is technically point identified.
- The AIC-preferred lower-order Bernstein specification $P = 4$ also gives tighter bounds than higher-order alternatives, which is useful for practice.

5.7 Figures 6 and 7: set-identified coverage for $Q = 4$ and $Q = 6$

Read:

- These figures study confidence sets for flexible functionals of utility in genuinely set-identified models.

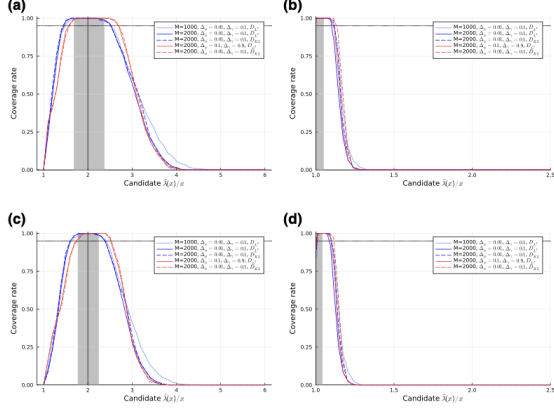


FIGURE 6

Flexible copula, $Q = 4$: Coverage (acceptance) rates based on level $\alpha = 0.95$ CSs for $\hat{\lambda}(x|y_0)/x$ via Procedure 2 at $x = 0.2$ and $x = 0.3$, parameterizing $\lambda_0(\cdot)$ as Bernstein of order $Q = 4$ and $F_1^0, \dots, F_K^0$ as Bernstein of order $P = 6$. (a) $\rho_0 = 0.5, x = 0.2$. (b) $\rho_0 = 0.0, x = 0.2$. (c) $\rho_0 = 0.5, x = 0.3$ and (d) $\rho_0 = 0.0, x = 0.3$.
Notes: Results based on 200 replications for Procedure 2-KL and 500 for Procedure 2- L^2 , with $B^* = 400$ parameter draws (ψ^B) per replication. Shaded areas are profiled $D_{KL}^0(\psi) \leq 10^{-6.0}$ contour sets. The vertical line is true $\hat{\lambda}(x|y_0)/x = 1/(1 - \rho_0)$.

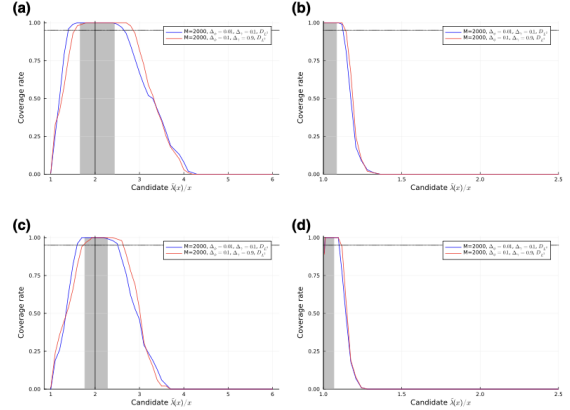


FIGURE 7

Flexible copula, $Q = 6$: Coverage (acceptance) rates based on level $\alpha = 0.95$ CSs for $\hat{\lambda}(x|y_0)/x$ via Procedure 2 at $x = 0.2$ and $x = 0.3$, parameterizing $\lambda_0(\cdot)$ as Bernstein of order $Q = 6$ and $F_1^0, \dots, F_K^0$ as Bernstein of order $P = 6$. (a) $\rho_0 = 0.5, x = 0.2$. (b) $\rho_0 = 0.0, x = 0.2$. (c) $\rho_0 = 0.5, x = 0.3$ and (d) $\rho_0 = 0.0, x = 0.3$.
Notes: Estimates based on 100 MC replications with $B^* = 400$ parameter draws each. Shaded areas are profiled $D_{KL}^0(\psi) \leq 10^{-6.0}$ contour sets. The vertical line is $\hat{\lambda}(x|y_0)/x = 1/(1 - \rho_0)$.

- The paper compares two implementations of Procedure 2: one using KL-based equivalence and one using an L^2 -style approximation implied by Theorem 1.
- The 95% confidence sets line up well with the targeted profiled-KL contour sets, which is exactly what the theory predicts once exact point identification is abandoned.
- The figures also show good power away from those contours. This is encouraging evidence that the procedure is useful in realistic set-identified equilibrium models, not just as a formal asymptotic device.

6 Short summary

- **Method.** The paper studies first-price auctions with risk-averse bidders and selective entry, derives a sharp nonparametric characterization of the bid-stage identified set, and develops robust likelihood-ratio inference for possibly set-identified models with parameter-dependent support.
- **Identification.** Risk aversion is identified from cross-environment restrictions linking bids, entry thresholds, and latent entrant-value quantiles. Risk neutrality is nonparametrically testable, and with enough variation in N and z the model primitives are identified up to a bounded constant.
- **Computation.** Flexible Bernstein parameterizations, MPEC with Chebyshev collocation, and stable profile-likelihood methods make estimation and inference practical even without closed-form bidding functions.
- **Main message.** Entry selection and lower-boundary conditions fundamentally change auction identification. Once those features are handled correctly, the paper shows that both sharp theory and usable finite-sample inference are possible.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a sharp set of equilibrium restrictions for first-price auctions with both risk aversion and selective entry.
- It shows that risk neutrality is testable and that sufficient variation in the number of potential bidders and in an entry-cost shifter identifies the primitives up to a bounded constant.
- It also shows that modern likelihood-ratio confidence-set methods can be adapted to flexible auction models with parameter-dependent support.

7.2 Economic intuition

- Selective entry means that the set of bidders who actually submit bids is endogenous. Therefore, changing competition changes both strategic bidding and the composition of bidder values.
- Risk aversion shows up through the inverse utility transformation λ^{-1} , which enters inverse bidding additively in quantile form.
- Cross-auction comparisons are informative because all auction environments must be reconciled with one common primitive environment. That is why the restrictions M, O, I, D, and S are so powerful.
- The reserve-price boundary condition matters because it determines whether the lower edge of the latent value support is pinned down or remains free up to an additive constant.

7.3 Takeaway

This paper is important both methodologically and substantively. Methodologically, it combines sharp equilibrium identification, flexible parameterization, and robust set-inference tools in an auction environment where support depends on the parameters. Substantively, it shows that the empirical study of bidder risk aversion cannot ignore selective entry or boundary conditions. In first-price auctions, these two ingredients are central to what can be learned from the data.